

From Gestural Handshapes MAN & WOMAN to Agreement: Pathways in Japanese Sign Language

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Revisiting MAN (TA) & WOMAN (I) and Grammatical Alignment

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Background & Our Goals

TA
(MAN/PERSON)

I
(WOMAN)



The TA and I handshapes have been used as gender-related gestures since at least the Edo period.

Osugi & Supalla (1998) provided a structural analysis of gender handshapes in Japanese Sign Language (JSL).

Since then, both usage patterns and grammatical environments have diversified.

Our goals

1. To reexamine the Ta/I system in JSL
2. To reorganize the pronoun system and its historical development
3. To compare data from the 1990s with the recent data (2024)

Previous Findings

Osugi & Supalla (1998) identified four major functions of gender handshapes in JSL:

1. Lexical use as *man* / *woman*, with pronominal extension
2. Gender marking within kinship terms, interacting with spatial hierarchy
3. Participation in agreement verb constructions
4. Use as classificatory markers in movement- and existence-centered predicates

👉 *Key claim:*

Gender handshapes contribute to agreement by spatially indexing subjects and objects.



OTA <TA> (TA)TELL+past(1), AMERICA <TA>+IX IX(2) MEET WANT,
(TA)TELL+past(1) IX(TA)

Ota told me, ' An American guy wants to meet you'.

Research Question

- What grammatical functions do TA/I co-occur with today?
- How do these functions differ between the 1990s and the 2020s?
- Are changes better captured through lexical categories or through syntactic alignment?

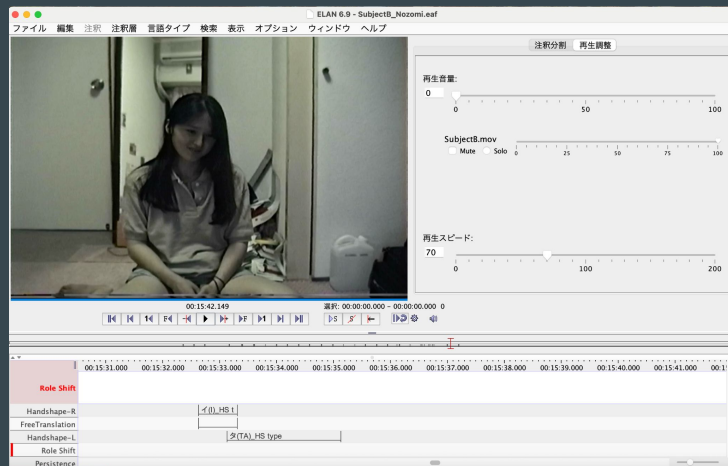
Data & Annotation

Our data consist of annotated video recordings from two periods: the 1990s and the 2020s.

Each token was annotated for handshape, referential function, movement, number emphasis, spatial case (alignment) such as Far or Near, kinship vocab, and politeness.

We also added tiers for subject–object relations, role shift, and alignment for both hands.

Importantly, the dataset primarily consists of utterances containing verbs.



Partial Annotation of the DATA

Referential functions are subdivided into two types:

- (i) pointing signs used for *he/she* in combination with the Ta or I handshape, and
- (ii) Ta or I handshapes accompanied by movement.

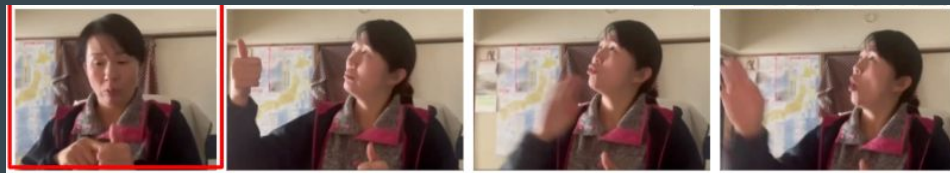
Referential
function



HE
SHE

Forefinger +referential

Uses Locus to specify meaning by context. Depending on verbs, the handshape TA and I is used continuously within the signing space. Verb type may change the case of the pronoun; possible use as relative pronoun



[MAN] [MAN] [ORDER]

(TA-handshape with IX) (TA-handshape) ORDER(b to a)

English: “The man orders from a staff (male) member”

Partial Annotation of the DATA

Referential functions are subdivided into two types:

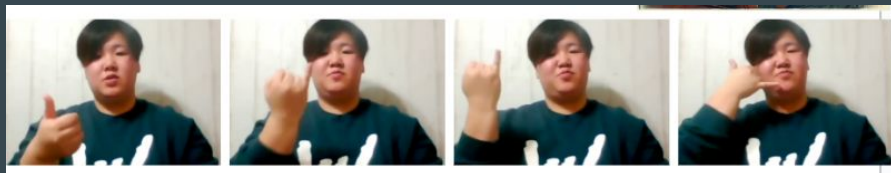
- (i) pointing signs used for *he/she* in combination with the Ta or I handshape, and
- (ii) Ta or I handshapes accompanied by movement.

Anaphoric
function with
movement



X's male
partner
X's female
partner

Functions as a possessive pronoun



[MAN] [WOMAN] [CALL]

(TA-handshpae) (I-handshape with movement)

English: “A man calls his (female) partner”

Overall Frequency

SUBJECT	TA	I	Total
A(2024)	21(48.2%)	25(51.8%)	46
B(1990)	27(48.2%)	29(51.8%)	56

Functional Features

Feature	Value	A (2024) (N=58)	B (1990)(N=52)
Movement Type	Yes	13	1
Number Emphasis	Singular	51	51
Alignment	Far	37	24
	Near	12	23
Kinship	No	51	51
Social Hierachy	No	52	49
Politeness adjustment	No	49	50

Observation Summary

Number, kinship, and politeness show largely identical distributions across the two speakers.

It should be noted that the data in this study were extracted primarily from verb-containing utterances, a factor that needs to be taken into account in interpreting these results.

Movement type is more frequent in Subject A(2024).

With respect to alignment, Subject A shows a clear bias toward *Far* values, indicating a preference for specific spatial case realizations.

The distribution of syntactic alignment also reveals an interesting asymmetry between the right and left hands.

Right vs Left Hand

- Subject B is left handed and uses both hands.
- Subject A is right-handed.

Subject B (1990):

- Both the amount and the range of functions are evenly distributed across the left and right hands.

Subject A (2024):

- The right hand shows functional concentration along with more diverse usage patterns.

Participants	Hands	I	TA	TOTAL
B	Left	14	15	29
B	Right	15	12	27
B Total		29	27	56
A	Left	2	4	6
A	Right	23	17	40
A Total		25	21	46

Gendered Handshapes, Dominant Hands, SA/OA in Relations

Far/Near values and subject–object roles do not correspond directly.

Rather, they are related as the outcome of syntactic alignment mediated by structural positioning.

A comparison of Participant A (2024) and Participant B (1990) shows the following patterns:

	Participant B (1990)	Participant A (2024)
Availability of SA/ OA	Appears on both hands	Appears on both hands
Stability of the right hand	High	Moderate
Consistency between syntactic roles and spatial placement	Highly consistent	Less consistent

Summary

In **JSL**, **case making and spatial configuration for agreement** often occur **simultaneously** , rather than as independent processes.

The notion of **alignment** became analytically salient only after systematic data organization. It is not always overtly visible without structured annotation.

Examining the relationship between **alignment** and **subject object relations** raises the question of whether traditional **verb type classifications in JSL** need to be reconsidered.

We also observe **persistence** , where spatial configurations are maintained, or reused across discourse.

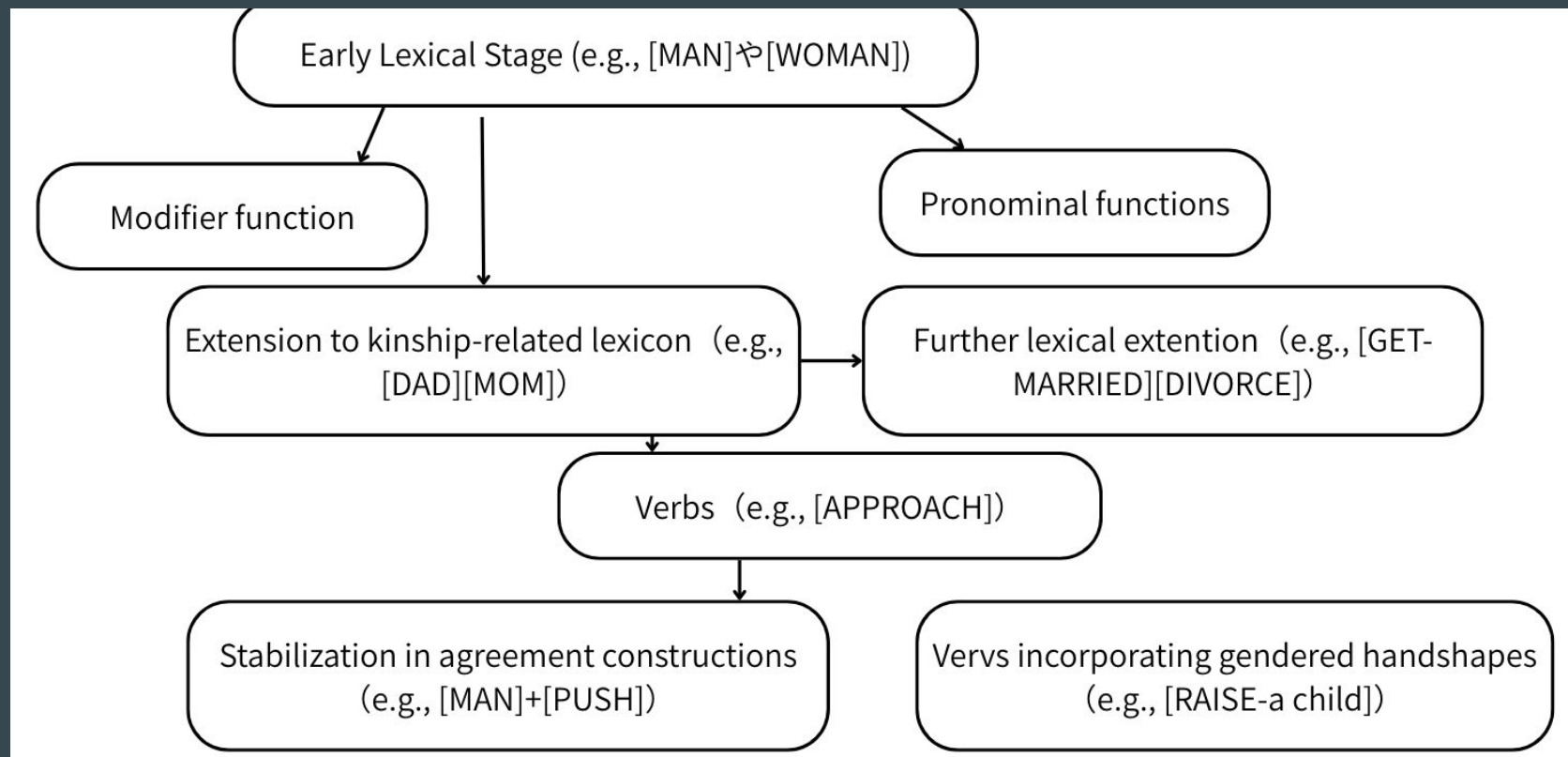
Taken together, agreement involving **TA and I handshapes** appears to be **discourse regulated** , rather than a fixed, verb-internal grammatical property.

Even apart from dominant-handedness, one hand may be preferentially used for organizing syntactic structure.

Appendix A: Subtypes of Pronominal functions in JSL

Classification	Handshape	Meaning	Function & Details
No Gender Marker		YOU I	Indicates deictically; no gender information
Referential function		HE SHE	Forefinger +referential Uses Locus to specify meaning by context. Depending on verbs, the handshape TA and I is used continuously within the signing space. Verb type may change the case of the pronoun; possible use as relative pronoun
Anaphoric function with movement		X's male partner X's female partner	Functions as a possessive pronoun
Plural emphasis		THEY WE	Plural and deictic and referential
Case Expression		[to X] [from X]	The closer location to the signer functions as the subjects, farther location as the object
Use in Kinship Contexts		[X's father] etc	Functions as a personal pronoun; grammatical case is determined by location. People of higher status are placed higher/gender information may be indicated/ pointing required before referencing kinship terms
Adjustment for Politeness and Respect		[Polite suffix]	Adjustments are made through spatial position and handshapes

Appendix B: functional pathway and agreement



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